Lecture-18
Differential Analysis of fluid motion II: Continuum mechanics to Newtonian models

Saikat Mukherjee

Department of Mechanical Engineering, IIT Kharagpur

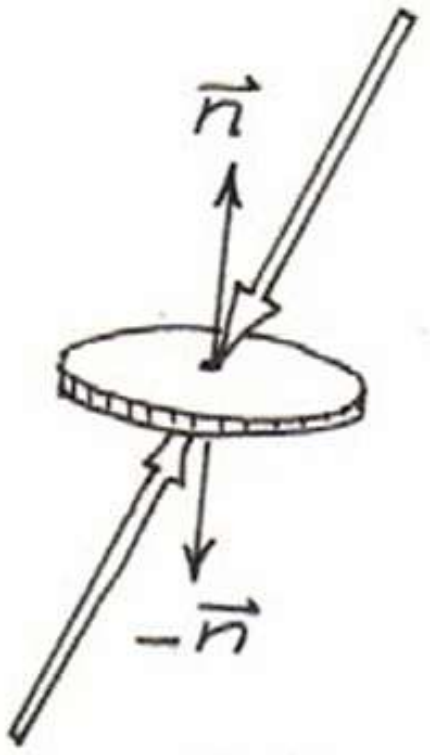
Learning Objectives

By the end of this lecture, you should be able to

1. Apply **Cauchy reciprocity** to prove how surface traction forces must balance out
2. Derive the **Cauchy stress tensor** from force balance on an infinitesimal tetrahedron
3. Explain the **symmetry of the Cauchy stress tensor** by applying the **conservation of angular momentum**
4. Formulate the **Newtonian constitutive model** for both incompressible and compressible fluids
5. Understand the motivation for **Stokes' hypothesis**

Cauchy reciprocity

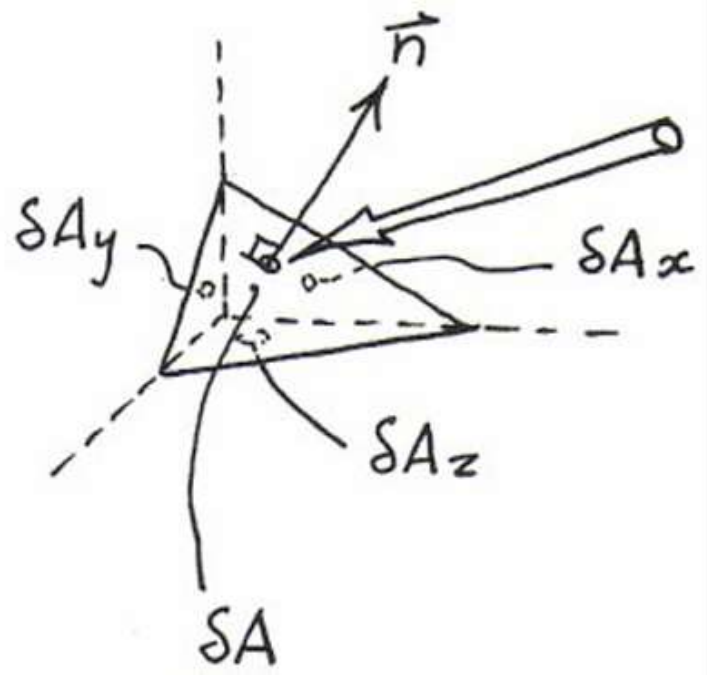
- **Traction** is a vector quantity that defines the **internal or external force per unit area** acting across an imaginary or real surface inside a continuous material.
- Because the **directional information is entirely intrinsic to the argument of the traction vector**, the force on an infinitesimal surface element is obtained simply by multiplying the traction vector by the scalar area.
- Traction vectors acting on opposite sides of the same internal surface at a given point are **equal in magnitude** and **opposite in direction**.



https://ocw.mit.edu/courses/2-25-advanced-fluid-mechanics-fall-2013/

Stress on an arbitrary plane

- The stress on an arbitrary plane is determined using Cauchy's relation, which states that the **traction vector** on a surface equals the **product of the stress tensor** and the **outward unit normal vector** of that plane.
- The **traction vector** depends directly on the **orientation of the cutting plane**.
- The **stress tensor** is a property of the material state at a point and **does not depend on the orientation of the surface.**
- The stress **transforms as a second-order tensor** under coordinate transformation.



Revisiting the learning objectives

1. Apply **Cauchy reciprocity** to prove how surface traction forces must balance out
2. Derive the **Cauchy stress tensor** from force balance on an infinitesimal tetrahedron
3. Explain the **symmetry of the Cauchy stress tensor** by applying the **conservation of angular momentum**
4. Formulate the **Newtonian constitutive model** for both incompressible and compressible fluids
5. Understand the motion for **Stokes' hypothesis**

